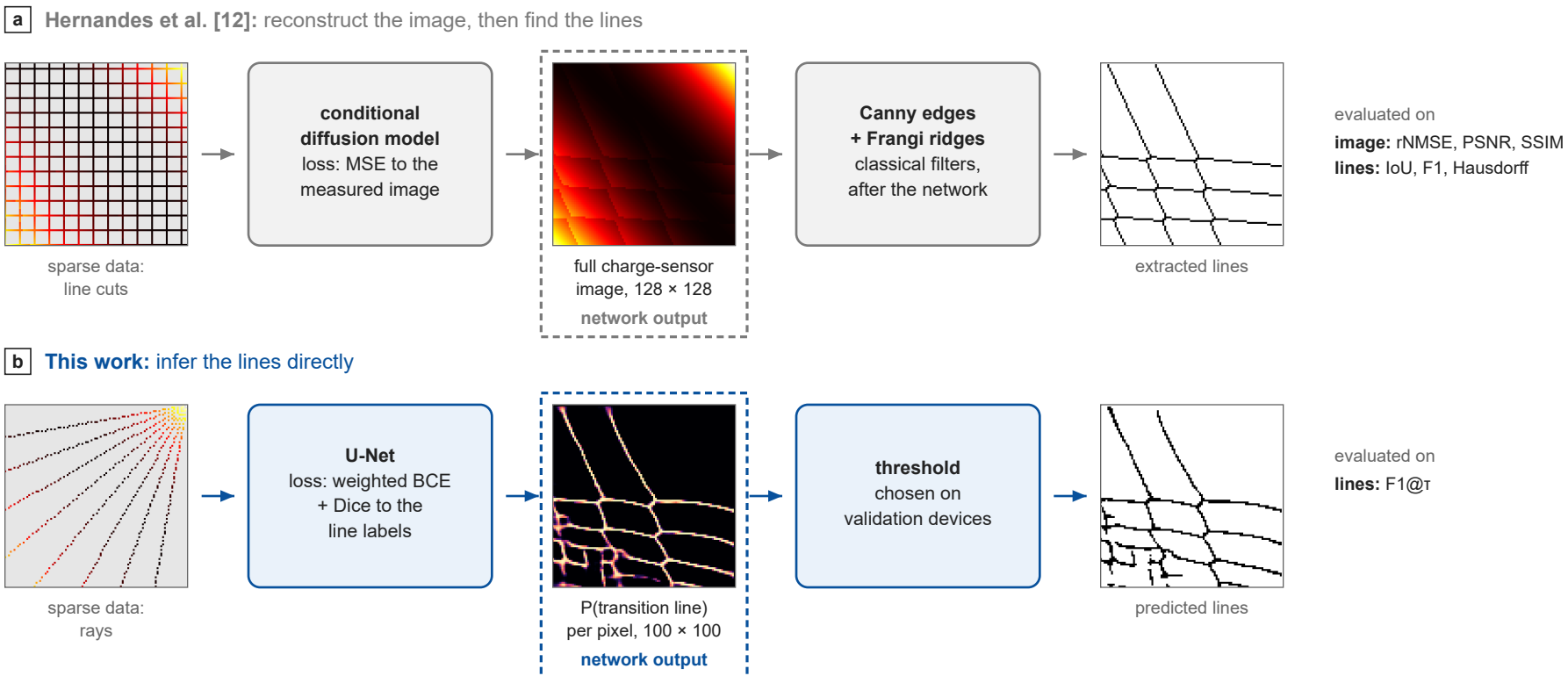

a Hernandes et al. [12]: reconstruct the image, then find the lines
sparse data: line cuts
conditional diffusion model
loss: MSE to the measured image
full charge-sensor image, 128 × 128
network output
Canny edges + Frangi ridges
classical filters, after the network
extracted lines
evaluated on
image: rNMSE, PSNR, SSIM
lines: IoU, F1, Hausdorff
b This work: infer the lines directly
sparse data: rays
U-Net
loss: weighted BCE + Dice to the line labels
P(transition line) per pixel, 100 × 100
network output
threshold
chosen on validation devices
predicted lines
evaluated on
lines: F1@τ